

Mach Zehnder Interferometer Design Proposal

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I. INTRODUCTION

THIS proposal discusses the design and simulation of a Mach-Zehnder Interferometer (MZI) fabricated in a photonic integrated circuit. The basic MZI splits input light into two arms, and then recombines them into one output. The phase of the light in the two arms when combining will determine the degree of constructive and destructive interference. The output can be detected using a conventional intensity detector. This is convenient, because most detectors cannot easily resolve phase change. However, by applying the MZI, phase can be converted into intensity information.

II. WAVEGUIDE

The waveguide was designed as a silicon (Si) rectangular channel in a silicon-dioxide (SiO₂) substrate. The Si channel was designed with dimensions 500 μm in width, and 0.22 μm in height. This is a standard selection which confines the mode well. The simulation results showing the transverse electric field (TE) and transverse magnetic field (TM) are shown in Fig. 1. It can be seen that the waveguide is TE polarized.

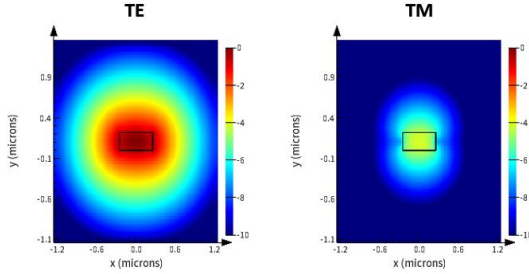


Fig. 1. TE and TM fields in the Si waveguide.

The effective index (n_{eff}) and group index (n_g) can also be simulated given a center wavelength (λ_0) of 1550 nm. The n_{eff} as a function of wavelength is shown in Fig. 2.

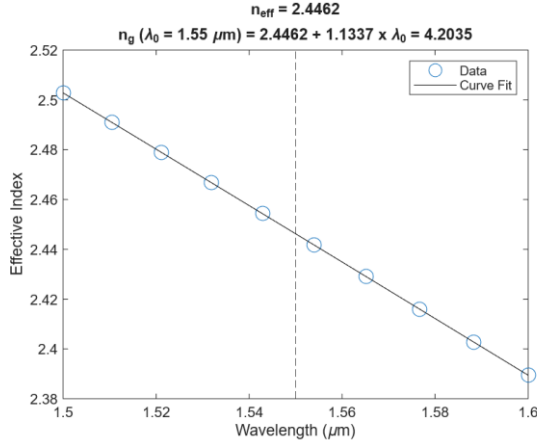


Fig. 2. n_{eff} as a function of wavelength, where the dotted vertical line marks the center wavelength.

However, since it is not convenient to re-run this simulation for any arbitrary wavelength, the compact model can be fitted. The compact model is essentially a polynomial fit to the data in the form of:

$$n_{\text{eff}}(\lambda) = n_1 + n_2(\lambda - \lambda_0) + n_3(\lambda - \lambda_0)^2 \\ = 2.446 - 1.134(\lambda - 1.55) - 0.044(\lambda - 1.55)^2$$

The effect of bending on TE mode confinement and transmission can also be modeled. A tight bend will result in poorer mode confinement and weaker transmission. This can be seen in Fig. 3, where a 2 μm bend has a significantly lopsided mode profile, while the 20 μm bend is much more symmetric. However, even with a relatively large bending radius, the number of bends can eventually lead to lead to appreciable bending losses. Also shown in Fig. 3 is the bending loss and transmission for the simulated fiber as a function of bend radius for 1 bend or 4 bends.

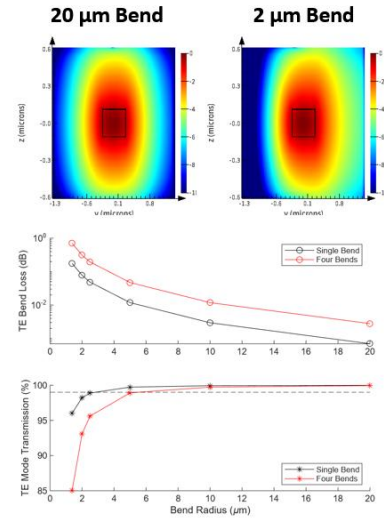


Fig. 3. Bending mode profiles, losses, and transmission for various bends radii. The profiles for a single and quad bends are shown as well. The horizontal dashed line signifies a transmission of 99%.

III. THE MACH-ZEHNDER INTERFEROMETER DESIGN

The imbalanced MZI has a transfer function described by:

$$T(\lambda) = \frac{I_0}{I_i} = \frac{1}{2} [1 + \cos(\beta \Delta L)] \\ \beta = \frac{2\pi n}{\lambda}$$

The free spectral range of the MZI is described by:

$$FSR = \frac{\lambda^2}{\Delta L(n_g)}$$

Thus, the length difference between the two MZI arms can be designed to target specific FSR widths. Table 1 shows the target FSR and calculated ΔL .

Design #	FSR (nm)	ΔL (μm)
1	5	114.3
2	10	57.2
3	15	38.1

Table 1. Three designs targeting different FSR widths and the associated calculated ΔL .

The gain profile for each design can be simulated using the waveguide as described in previous section, Y-branches, and grating couplers. Shown in Fig. 4, it can be seen that the periodicity of the gain profile decreases as the FSR increases.

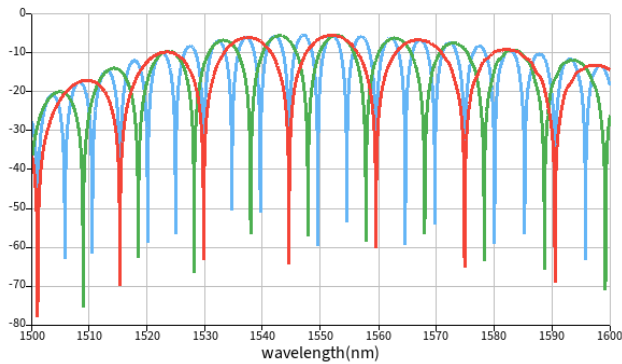


Fig. 4. Gain profiles for Design #1 (blue), Design #2 (green), and Design #3 (red).

The simulated FSR is also shown in Fig. 5 to check how well it matches the theoretical calculations.

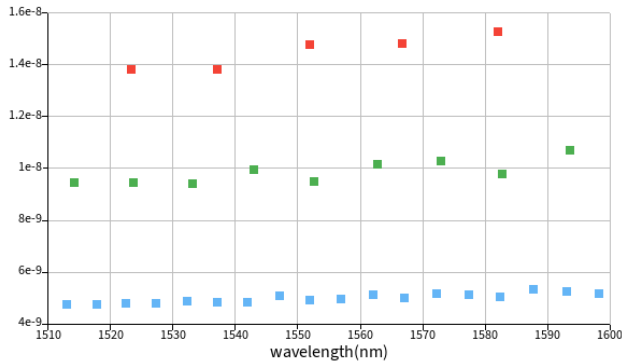


Fig. 5. Simulated FSR as a function of wavelength.

Finally, the group index can be calculated by solving for ng using the FSR equation.

$$ng = \frac{\lambda^2}{\Delta L(FSR)}$$

This calculation can be applied to simulation data, which directly displays FSR. Fig.5 shows the calculated group index for Design #1. It can also be applied to measured data. If the measured data does not automatically display FSR, one would need to find the fringe peaks and calculate the distance between the peaks in post-processing.

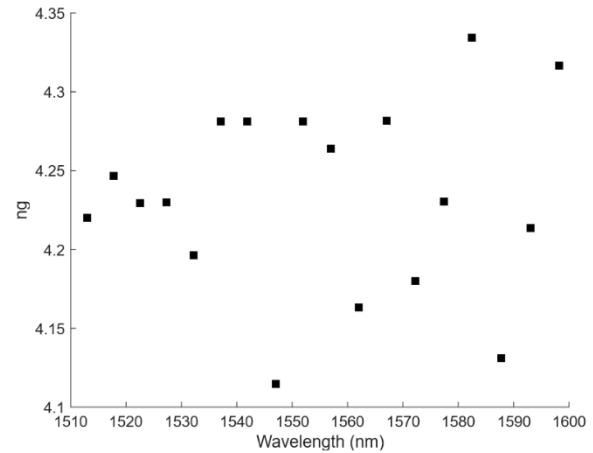


Fig. 5. ng calculated from the simulated FSR data of Design #1.

The layout for the MZI circuits are shown in Fig. 6. Four designs are shown from left to right starting with a balanced MZI, and increasing ΔL as defined by Table 1.

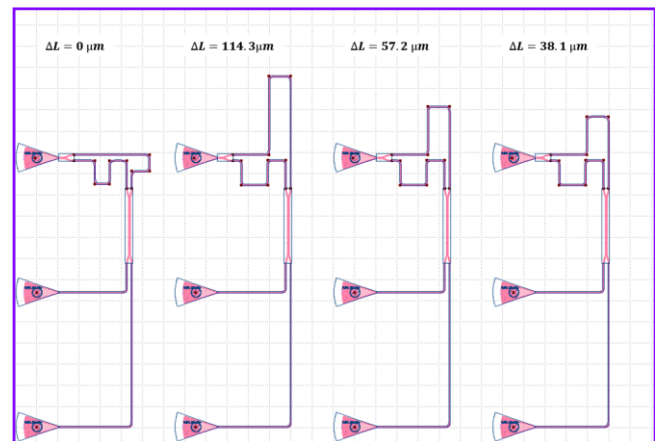


Fig. 6. Layouts for fabrication are shown from left to right: Balanced MZI ($\Delta L = 0$), Design 1, Design 2, and Design 3, as defined by Table 1.